RAJA M
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Internship

Intern | Travancore Titanium Products Limited (TTPL), Trivandrum **Jun 2025 – Jul 2025**

- Exposure to sulphate-process-based TiO_2 manufacturing and industrial quality control practices.
- Analyzed sulphate process flow for TiO_2 production and studied reaction kinetics involved in hydrolysis and calcination stages.
- Learned process flow diagrams (PFDs), equipment layouts, and process control basics.
- Examined plant safety systems, waste treatment processes, and energy management practices.

Student Volunteer | Community Development Project, Karur **July 2024**

- Studied sustainable farming practices with an emphasis on organic methods and soil health.
- Introduced and demonstrated seed coating technology to local farmers for better crop yield.
- Explored participatory development models that link agriculture with social welfare

Projects

Biodegradable Smart Packaging Film with Dual Antimicrobial Action **Oct 2025 – Dec 2025**

- Applied Chemical Reaction Engineering principles to develop a biodegradable smart packaging film with dual antimicrobial functionality.
- Analyzed reaction kinetics and diffusion-driven mass transfer governing controlled antimicrobial release.
- Designed a polymer-based antimicrobial system for sustained activity and material stability.

Automated Water Level Detection System **Mar 2024 – Apr 2024**

- Built a sensor-based system for real-time liquid level monitoring.
- Integrated sensors, circuits, and Arduino for automation.
- Calibrated system to improve measurement accuracy.

Design & Fabrication of Laboratory-Scale Batch Reactor **Oct 2023 – Dec 2025**

- Designed and fabricated a lab scale batch reactor for controlled chemical reactions.
- Conducted experimental testing and calibration to evaluate temperature distribution and reaction stability.
- Selected materials based on corrosion resistance, thermal stability, and chemical compatibility.
- Key Learning: Practical understanding of heat transfer limitations and reaction rate control in closed systems.

Certificates

Digital Marketing Course **Dec 2025**
Udemy

Introduction to HTML and CSS **April 2024**
Mind Luster

Skills

Technical Skills: Reaction Engineering, Heat Transfer, Mass transfer, Process Control, Experimental data analysis.

Software:MATLAB, ANSYS, Creo, Canva, Excel.

Soft Skills: Problem solving, Communication Skills, Critical Thinking, Teamwork & Collaboration.

Programming: Python, C++ (Basic), HTML, CSS

Education

Lovely Professional University Aug **2023 – Present**
Bachelor of Technology Chemical Engineering — CGPA:7.58 *Phagwara, Punjab*

Velammal Bodhi Campus **2022 – 2023**
12th — Percentage: 87% *Karur, Tamil Nadu*

Velammal Bodhi Campus **2020 – 2021**
10th — Percentage: 90% *Karur, Tamil Nadu*